Spectral Clustering

① Similarity graphs

Let $x_1, \dots, x_n$ be n observations

and $s_{i,j} \geq 0$ the similarity between x_i and x_j

The goal of clustering is to divide the observations into groups, such that points in the same group are similar to each other and points in $\neq$ groups are dissimilar to each other.

Let $G := (V, E)$ be a similarity graph representing the similarities between $(x_1, \dots, x_n)$

$V = (v_1, \dots, v_n)$ vertices, $\forall i \in [\![1, n]\!]$, v_i represents vertex of data point x_i

x_i, x_j are connected if $s_{i,j} \geq t$) some threshold

$s_{i,j}$: weight of the edge between v_i and v_j

Goal: Clustering $x_1, \dots, x_n$, in terms of similarity graph can be reformulated :

Find a partition of the graph such that edges between $\neq$ groups have very low weights (i.e. dissimilar to each other) and edges in the same groups have very high weights (i.e very similar to each other).

1.1. Notations

Let $G = (V, E)$ be an undirected weighted graph

with vertex $V = \{v_1, \dots, v_n\}$

and weights $W = (w_{i,j})_{1 \leq i,j \leq n} \in \mathcal{M}_n(\mathbb{R})$

and such that $\forall i, j \quad w_{ij} \geq 0$

W is the weighted adjacency matrix of the graph

$\forall i, j \quad w_{i,j} = 0$ i.e. v_i and v_j are not connected

G is undirected therefore $\forall i, j \quad w_{ij} = w_{ji}$

hence $G^T = G$ i.e. G is symmetric

We define the degree of a vertex $v_i \in V$ such that

$$d_i = \sum_{j=1}^{n} w_{i,j}$$

The sum of all weights of all edges adjacent to vertex v_i.

We call the degree matrix

$$D := \operatorname{diag}(d_1, \dots, d_n) = \begin{bmatrix} d_1 & & 0 \\ & \ddots & \\ 0 & & d_n \end{bmatrix}$$

Let $A \subset V$ a subset of vertices

- $\bar{A} = V \setminus A$ its complement
- $\mathbb{1}_A = (f_1, \dots, f_n)' \in \mathbb{R}^n$: indicator vector of A

  $\forall i \in [\![1, n]\!]$, $f_i = 1$ if $v_i \in A$ else 0
- $\{i \in A\} = \{i \in [\![1, n]\!] \mid v_i \in A\}$
- $\forall A, B$ subsets of V, we define

$$W(A, B) = \sum_{i \in A, j \in B} w_{i,j}$$

- Two ways of measuring the size of a subset $A \subset V$

$$|A| := \text{number of vertices in } A$$

$$\operatorname{vol}(A) := \sum_{i \in A} d_i$$

- $A \subset V$ is said connected if $\forall v_i, v_j \in A$ they can be joined by a path trough A
- $A \subset V$ is called a connected component if it is connected and there are no connection between A and $\bar{A}$
- $A_1, \dots, A_k$ is a partition of the graph if

  $\forall i, j \in [\![1, k]\!]$, $A_i \cap A_j = \phi$

  and $\displaystyle\bigsqcup_{i=1}^{k} A_i = V$

1.2. Different similarity graphs

$$x_1, \dots, x_n \longrightarrow \begin{array}{l} s_{i,j} : \text{similarities} \\ d_{i,j} : \text{distances} \end{array}$$

goal: model local neighborhood relationships between data points.

ε-neighborhood graph:

$$s_{i,j} = \begin{cases} 1 & \text{if } d(x_i, x_j) < \varepsilon \\ 0 & \text{else} \end{cases}$$

k-nearest neighbor graphs

connect v_i to v_j if v_j is in the top k-neighbors of v_i

⚠ to keep the graph undirected:

↳ ignore the direction of the edge : k-nearest neighbor graph

↳ connect the vertex if both are in top k neighbors of the other : mutual k-nearest neighbor graph

fully connected graph:

example $\quad s_{i,j} = \exp\left(-\dfrac{\|x_i - x_j\|^2}{2\sigma^2}\right)$

② Graph Laplacians

⚠ There is no unique convention which matrix is called « graph Laplacian »

In the following we assume G undirected, weighted with weight matrix $W = (w_{i,j})_{1 \leq i,j \leq n}$ and $\forall i,j \ w_{i,j} \geq 0$

Referring to an eigen vector we will assume it is normalized.

2.1. Unnormalized graph Laplacian

$$L := D - W$$

Proposition 1:

a) $\forall f \in \mathbb{R}^n, \quad f^T L f = \frac{1}{2} \sum_{i,j} w_{i,j}(f_i - f_j)^2$

b) L is symmetric and positive semi-definite

c) $\min \operatorname{spec}(L) = 0$, the corresponding eigenvector is the constant vector $\mathbb{1}_n$

d) L has non négative, real-valued eigenvalues

$$0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$$

Proof:

a) Let $f \in \mathbb{R}^n$

$$f^T L f = f^T(D-W)f = f^T D f - f^T W f$$

$$= \sum_{i=1}^{n} d_i f_i^2 - \sum_{1 \leq i,j \leq n} w_{i,j} f_i f_j$$

$$= \frac{1}{2}\sum_{i=1}^{n} d_i f_i^2 + \frac{1}{2}\sum_{j=1}^{n} d_j f_j^2 - \sum_{1 \leq i,j \leq n} w_{i,j} f_i f_j$$

$$= \frac{1}{2}\left[\sum_{i=1}^{n} d_i f_i^2 - 2\sum_{1 \leq i,j \leq n} w_{i,j} f_i f_j + \sum_{j=1}^{n} d_j f_j^2\right]$$

D definition

$$= \frac{1}{2}\left[\sum_{1 \leq i,j \leq n} w_{i,j} f_i^2 - 2\sum_{1 \leq i,j \leq n} w_{i,j} f_i f_j + \sum_{1 \leq i,j \leq n} w_{i,j} f_j^2\right]$$

$$= \frac{1}{2}\sum_{1 \leq i,j \leq n} w_{i,j}\left(f_i^2 - 2 f_i f_j + f_j^2\right)$$

$$f^T L f = \frac{1}{2}\sum_{1 \leq i,j \leq n} w_{i,j}(f_i - f_j)^2$$

b) $L^T = (D-W)^T = D^T - W^T = \underbrace{D}_{\text{diagonal}} - \underbrace{W}_{\text{symmetric}} = L$

c) cf definition of L and D.

d) $\left.\begin{array}{l} a) \Rightarrow L \geq 0 \\ b) \ L \text{ is symmetric} \end{array}\right) \rightarrow$ cf Spectral Theorem

Remarks

Two matrices with the same coefficients outside the diagonal but with ≠ diagonal coefficients have the same unnormalized graph Laplacian L.

i.e. self edges in a graph do not change the corresponding Laplacian.

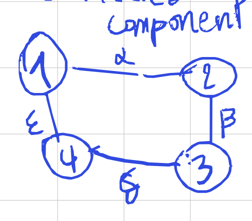
Proposition 2 (Number of connected components and the spectrum of L)

Let G be an undirected graph with non-negative weights. Then the multiplicity k of the eigenvalue 0 of L equals the number of connected components $A_1, \dots, A_k$ in the graph.

The eigenspace of eigenvalue 0 is spanned by the indicator vectors $\mathbb{1}_{A_1}, \dots, \mathbb{1}_{A_k}$ of those components.

Examples

1 connected component

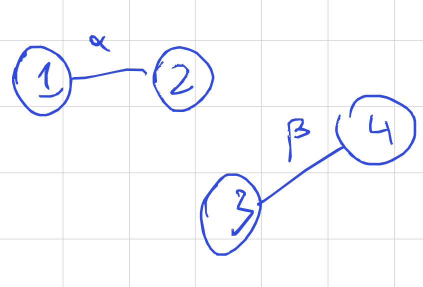


$$W = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & \alpha & 0 & \varepsilon \\ 2 & \alpha & 0 & \beta & 0 \\ 3 & 0 & \beta & 0 & \delta \\ 4 & \varepsilon & 0 & \delta & 0 \end{array} \quad ; \quad D = \begin{bmatrix} \alpha+\varepsilon & 0 & 0 & 0 \\ 0 & \alpha+\beta & 0 & 0 \\ 0 & 0 & \beta+\delta & 0 \\ 0 & 0 & 0 & \varepsilon+\delta \end{bmatrix}$$

$$L = \begin{bmatrix} \alpha+\varepsilon & -\alpha & 0 & -\varepsilon \\ -\alpha & \alpha+\beta & -\beta & 0 \\ 0 & -\beta & -(\beta+\delta) & -\delta \\ -\varepsilon & 0 & -\delta & -(\varepsilon+\delta) \end{bmatrix}$$

$$L\mathbb{1}_n = \begin{bmatrix} \quad \end{bmatrix}\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{i.e. } L\mathbb{1} = 0$$

2 connected components



$$W = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & \alpha & 0 & 0 \\ 2 & \alpha & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 & \beta \\ 4 & 0 & 0 & \beta & 0 \end{array}$$

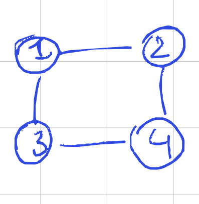
$$L = \begin{bmatrix} \alpha & -\alpha & 0 & 0 \\ -\alpha & \alpha & 0 & 0 \\ 0 & 0 & +\beta & -\beta \\ 0 & 0 & -\beta & \beta \end{bmatrix} \qquad \mathbb{1}_{A_1} = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} ; \ \mathbb{1}_{A_2} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}$$

$$E_0(L) = \operatorname{Vect}(\mathbb{1}_{A_1}, \mathbb{1}_{A_2})$$

Proof:

Let $k := 1$ (number of connected components)
i.e. the graph is connected

[HQ $E_0(L) = \operatorname{span}(\mathbb{1}_n) = \operatorname{span}(\mathbb{1}_V)$



Let $f \in E_0(L)$ i.e. $Lf = 0 \cdot f = 0$

so $f^T L f = \sum_{1 \leq i,j \leq n} w_{i,j}(f_i - f_j)^2 = 0$

$$= \sum_{\substack{1 \leq i,j \leq n \\ w_{i,j} \neq 0}} \underbrace{w_{i,j}}_{\geq 0} \underbrace{(f_i - f_j)^2}_{\geq 0} = 0$$

so $\forall i,j, \quad w_{i,j} = 0$ or $f_i = f_j$

Let $i \in [\![1, n]\!]$, the whole graph is connected

so there exist $j \in [\![1, n]\!], j \neq i$ such that $w_{i,j} > 0$

so $f_i = f_j$

$(\dots) \quad \forall i,j \quad f_i = f_j = \alpha \in \mathbb{R}$?!

donc $f = \alpha \mathbb{1}_n \in \operatorname{span}(\mathbb{1}_n) = \operatorname{span}(\mathbb{1}_V)$

Case $k > 1$

i.e. there exists k connected components $A_1, \dots, A_k$.

Let us suppose the vertices ordered according to the connected components i.e.

$$W = \begin{bmatrix} W_1 & & 0 \\ & \ddots & \\ 0 & & W_k \end{bmatrix} \quad \text{with } W_1, \dots W_k \text{ blocks}$$

Therefore
$$L = \begin{bmatrix} L_1 & & 0 \\ & \ddots & \\ 0 & & L_k \end{bmatrix} \quad \text{with } L_1, \dots, L_k \text{ blocks}$$

L_i is a the Laplacian of a connected graph so (cf previous)

$E_0(L_i) = \operatorname{span}(\mathbb{1}_{A_i})$ in $\mathbb{R}^{n_i}$

$(\dots)$

So $E_0(L) = \operatorname{span}(\mathbb{1}_{A_1}, \dots, \mathbb{1}_{A_k})$

so the multiplicity of 0 as eigen value of L is the number of connected components

2.2. Normalized graph Laplacians

$L := D - W$

$$L_{sym} := D^{-1/2} L D^{-1/2} = I - D^{-1/2} W D^{-1/2}$$

$$L_{rw} := D^{-1} L = I - D^{-1} W$$

L_{sym} : cf symetric

L_{rw} : cf closely related to a random walk

Proposition 3

1) $\forall f \in \mathbb{R}^n, \quad f^T L_{sym} f = \frac{1}{2} \sum_{1 \leq i,j \leq n} w_{ij} \left(\frac{f_i}{\sqrt{d_i}} - \frac{f_j}{\sqrt{d_j}} \right)^2$

2) $u \in E_\lambda(L_{rw}) \iff D^{1/2} u \in E_\lambda(L_{sym})$

3) $u \in E_\lambda(L_{rw}) \iff Lu = \lambda D u$

4) $\mathbb{1} \in E_0(L_{rw}) \; ; \; D^{1/2} \mathbb{1} \in E_0(L_{sym})$

5) $L_{sym} \geq 0 \; , \; L_{rw} \geq 0$

$\quad Spec(L_{sym}) \subset \mathbb{R}^+ \; ; \; Spec(L_{rw}) \subset \mathbb{R}^+$

$\quad 0 \in Spec(L_{sym}) \; ; \; 0 \in Spec(L_{rw})$

Proof:

1) $L_{sym} = D^{-1/2} L D^{-1/2}$ so $f^T L_{sym} f = \underbrace{f^T D^{-1/2}}_{u^T} L \underbrace{D^{-1/2} f}_{u}$

$$= \sum_{1 \leq i,j \leq n} w_{ij} (u_i - u_j)^2$$

$u = D^{-1/2} f$ donc $\forall i, \; u_i = \frac{f_i}{\sqrt{d_i}}$

2) $u \in E_\lambda(L_{rw}) \iff u \neq 0, \; L_{rw} u = \lambda u$

$\iff u \neq 0, \; D^{-1} L u = \lambda u$

$\iff u \neq 0, \; D^{-1} L D^{-1/2} D^{1/2} u = \lambda u$

$\iff u \neq 0, \; D^{1/2} D^{-1} L D^{-1/2} (D^{1/2} u) = \lambda D^{1/2} u$

$\iff D^{1/2} u \neq 0, \; D^{-1/2} L D^{-1/2} (D^{1/2} u) = \lambda D^{1/2} u$

$\iff D^{1/2} u \neq 0 \quad L_{sym}(D^{1/2} u) = \lambda D^{1/2} u$

$\iff D^{1/2} u \in E_\lambda(L_{sym})$

3) $u \in E_\lambda(L_{rw}) \iff u \neq 0, \; L_{rw} u = \lambda u$

$\iff u \neq 0, \; D^{-1} L u = \lambda u$

$\iff u \neq 0, \; D D^{-1} L u = \lambda D u$

$\iff u \neq 0, \; L u = \lambda D u$

4) $L \mathbb{1} = 0$ so $D^{-1} L \mathbb{1} = 0$ so $L_{rw} \mathbb{1} = 0$

so $\mathbb{1} \in E_0(L_{rw})$

② $\Rightarrow D^{1/2} \mathbb{1} \in E_0(L_{sym})$

5) • 1) $\Rightarrow L_{sym} \geq 0$, L_{sym} symmetric

so $spec(L_{sym}) \subset \mathbb{R}^+$

$L_{sym} \mathbb{1} = 0 \cdot \mathbb{1}$ donc $0 \in Spec(L_{sym})$

* 1) $\Rightarrow spec(L_{rw}) = Spec(L_{sym}) \subset \mathbb{R}^+$

et $0 \in spec(L_{rw})$

Proposition 4

(Number of connected components and spectra of L_{sym} and L_{rw})

Let G be an undirected graph with non negative weights.

Then the multiplicity k of the eigenvalue 0 of both L_{rw} and L_{sym} equals the number of connected components $A_1, \ldots, A_k$ in the graph.

$$E_0(L_{rw}) = span(\mathbb{1}_{A_1}, \ldots, \mathbb{1}_{A_k})$$

$$E_0(L_{sym}) = span(D^{1/2} \mathbb{1}_{A_1}, \ldots, D^{1/2} \mathbb{1}_{A_k})$$

Proof: cf Prop 2 & Prop 3

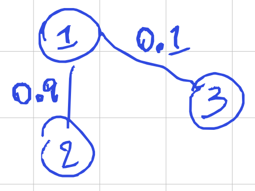
③ Spectral Clustering Algorithms

Let $x_1, \dots, x_n$ be n observations

$s(x_i, x_j) = s_{i,j} \geq 0$ their similarities

such that $\forall i,j \quad s_{i,j} = s_{j,i}$

$S := (s_{ij})_{1 \leq i,j \leq n}$: similarity matrix



$$S = \begin{array}{c|ccc} & 1 & 3 & 2 \\ \hline 1 & 0 & 0.1 & 0.9 \\ 3 & 0.1 & 0 & 0 \\ 2 & 0.9 & 0 & 0 \end{array}$$

$$W_G = \begin{bmatrix} 0 & 0.1 & 0.9 \\ 0.1 & 0 & 0 \\ 0.9 & 0 & 0 \end{bmatrix} \qquad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad L = \begin{bmatrix} 1 & -0.1 & -0.9 \\ -0.1 & 0.1 & 0 \\ -0.9 & 0 & 0. \end{bmatrix}$$

$$\text{spec}(L) = \{0, \overset{\simeq 0}{\overbrace{0.1456}}, 1.8544\}$$

↳ eigenvectors $u_1, \dots, u_k$

Unnormalized spectral clustering

Input: S : similarity matrix, k : number of clusters

- Construct similarity graph $G = (V, E)$
  compute W, D, L

(2)
- Compute the first k eigenvectors $u_1, \dots, u_k$ of L (associated to lowest k eigenvalues)
- $U = \begin{pmatrix} | & & | \\ u_1 & \cdots & u_k \\ | & & | \end{pmatrix} = \begin{pmatrix} — & y_1 & — \\ & \vdots & \\ — & y_n & — \end{pmatrix} \in \mathcal{M}_{n,k}(\mathbb{R})$

Cluster points $(y_i)_{1 \leq i \leq n}$ (with $\forall i \ y_i \in \mathbb{R}^k$)

with k-means into clusters $C_1, \dots, C_k$

Output: Clusters $A_1, \dots, A_k$ with

$$A_i = \{ j \in [\![1, n]\!] \mid y_j \in C_i \}$$

Normalized spectral clusterings

① for (2) compute $u_1, \dots, u_k$ such that

$$L u_j = \lambda D u_j$$

② ↳ Use L_{sym}
↳ Compute $u_1, \dots, u_k$ eigen vectors of L_{sym}

$$U = \begin{pmatrix} | & & | \\ u_1 & \cdots & u_k \\ | & & | \end{pmatrix} = \begin{pmatrix} — & \tilde{y}_1 & — \\ & \vdots & \\ — & \tilde{y}_n & — \end{pmatrix}$$

$$T = \begin{pmatrix} — & y_1 & — \\ & \vdots & \\ — & y_n & — \end{pmatrix} \quad \text{with } y_i = \frac{1}{\|\tilde{y}_i\|} \tilde{y}_i$$

Clustering with k-means of $(y_1, \dots, y_n)$